

Yixiong Sun

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PROFESSIONAL SUMMARY

Systems neuroscience PhD candidate and computational researcher with expertise in neural engineering, chronic *in vivo* electrophysiology, machine learning, signal processing, and mechanistic modeling. Leads end-to-end investigations of neural circuits and memory, combining custom recording and closed-loop stimulation systems, freely moving animal experiments, large-scale neural data analysis, and quantitative modeling. Builds on prior production software engineering experience to design and integrate complex experimental and computational systems.

EDUCATION

University of Toronto Toronto, ON
PhD, Cell and Systems Biology (Systems Neuroscience) Sep 2021 – Present
Thesis: Linking abnormal hippocampal neural activity and memory deficits to A β accumulation in Alzheimer's disease
GPA: 4.0/4.0

McGill University Montreal, QC
Bachelor of Science, Joint Honours Computer Science and Biology Sep 2017 – Apr 2021
GPA: 3.96/4.0

EXPERIENCE

Takehara Lab, University of Toronto Toronto, ON
PhD Candidate, Systems Neuroscience Sep 2021 – Present
Data Sciences Institute Doctoral Student Fellowship (2024–Present) · \$25,000/year · 14 recipients

- Led a multi-year investigation of hippocampal network dysfunction in Alzheimer's disease, identifying reduced sharp-wave ripple activity and disrupted experience-dependent neuronal ensemble reorganization associated with memory deficits.
- Designed and conducted longitudinal hippocampal single-unit and LFP recordings in freely moving mice, including adjustable microdrive fabrication, stereotaxic implantation, and multi-day behavioral and sleep experiments.
- Designed and conducted spatial-memory and peptide-treatment experiments in an Alzheimer's mouse model, demonstrating improved memory performance and increased hippocampal ripple activity following treatment.
- Engineered and validated closed-loop neural stimulation systems for ripple-triggered cortical slow-wave induction and theta-phase-locked stimulation to investigate oscillatory coupling, synaptic plasticity, and memory impairment.
- Developed Python and MATLAB pipelines to process ~4 TB of electrophysiology, behavioral, and imaging data from 30+ mice, automating neural-event detection, population analyses, statistical testing, and visualization.
- Collaborated to extend a CA3–CA1 spiking network model, testing synaptic mechanisms of hippocampal dysfunction against experimental recordings.
- Mentored nine undergraduate researchers on independent projects and experimental work; trained graduate students across labs in electrophysiology techniques and data analysis.

Waldispühl Lab, McGill University Montreal, QC
Undergraduate Researcher Sep 2020 – Apr 2021

- Extended and validated BayesPairing2, an RNA secondary-structure and motif discovery pipeline, optimizing Bayesian-network construction and RNA model loading.

Beagle Inc Kitchener, ON
Software Developer May 2021 – Aug 2021

- Improved PDF/Word parsing and machine-learning pipelines for legal phrase recognition, incorporating supervised classifiers and an iterative data-labeling workflow.
- Refactored and containerized a legacy Django platform, implementing CI/CD pipelines for deployment to load-balanced AWS EC2 instances with S3 storage and PostgreSQL on Amazon RDS.

iVenuto.com Corporation Kitchener, ON
Software Developer, Consultant May 2018 – Aug 2021

- Owned the technical architecture and full-stack development of Zap In, a visitor-management platform serving 100+ customers, spanning JavaScript/Node.js, MongoDB, Redis, and iOS/WebView integration.

- Developed WebRTC video calling, offline synchronization, speech recognition, and authentication features. Overhauled the login experience, replacing static forms with an interactive, step-by-step interface.
- Engineered CI/CD pipelines and managed AWS-hosted staging and production environments, including Linux/Nginx servers, databases, background workers, and SSL/TLS configuration; delivered 10+ software releases.
- Led feature planning and code reviews for approximately two interns per term; investigated and resolved complex production issues escalated from customer support.

GHP Group Inc.

Mobile Software Developer (Contract)

Remote

Jan 2016 – Dec 2016

- Developed and released iOS and Android applications for Bluetooth Low Energy (BLE) devices, implementing device pairing and bidirectional command-and-response communication.

SELECTED TECHNICAL PROJECTS

EvidNeuro – Evidence-Grounded Neuroscience Retrieval Engine

Aug 2026 – Present

TypeScript, LLMs, MinerU, PostgreSQL/pgvector

- Designed a layered extraction pipeline that transforms neuroscience papers into linked source passages, scientific claims, measurements, biological entities, and knowledge-graph relationships, retaining direct links to the original text.
- Built multi-layer retrieval combining BM25/vector search, LLM reranking, and knowledge-graph queries, enabling cross-paper evidence discovery and direct navigation from extracted findings to supporting source passages.
- Processed 60 neuroscience papers into 15,000+ evidence objects and 9,000+ structured claims, supported by 1,000+ passing automated tests.

Hippocampal Event Classification

Jan 2026 – Present

Python, PyTorch, signal processing, CNNs, subject-aware evaluation

- Developed an end-to-end pipeline to detect and distinguish physiological hippocampal ripples from epileptiform discharges and noise in heterogeneous LFP recordings, using 7,701 labeled events from 30 subjects.
- Designed a multi-representation CNN combining multichannel waveforms, time-frequency spectrograms, cross-channel attention, and coherence- and entropy-based features to capture differences in event morphology and oscillatory structure.
- Achieved 0.849 macro F1 and 86.1% accuracy in subject-grouped cross-validation (5 folds $\times$ 3 seeds), outperforming an RBF-SVM baseline (0.753 macro F1, 77.1% accuracy) in all 15 paired evaluations.

Mechanistic Modeling of Hippocampal Network Dysfunction

Sep 2024 – Dec 2025

Python, Spiking Neural Networks, Computational Neuroscience, Brian2

- Extended a CA3–CA1 spiking network model to simulate experience-dependent synaptic plasticity and ripple-associated neuronal recruitment, reproducing changes in hippocampal activity following learning.
- Tested excitatory and inhibitory circuit perturbations against experimental recordings from Alzheimer’s mouse models, finding that increased synaptic excitation better reproduced impaired neuronal recruitment and ensemble reorganization.

TECHNICAL SKILLS

Programming: Python, TypeScript/JavaScript, MATLAB, C/C++, Bash

Machine Learning and AI: PyTorch, scikit-learn, CNNs, transfer learning, model evaluation, local LLM inference (Qwen), LLM-assisted extraction, hybrid retrieval (BM25/vector search), knowledge graphs

Scientific Computing: NumPy, SciPy, pandas, digital signal processing, time-frequency analysis, statistical analysis, graph/network analysis, spiking neural networks (Brian2)

Software and Cloud: Node.js, Django, PostgreSQL, MongoDB, Redis, Docker, AWS (EC2/S3), CI/CD, Linux, Nginx, Git

Neuroscience and Neural Engineering: Chronic single-unit/LFP recordings, microdrive fabrication and implantation, closed-loop stimulation, neural recording systems (Neuralynx, Open Ephys), Arduino, hardware/software integration, behavioral experiments

SELECTED PUBLICATIONS AND PRESENTATIONS

Sun Y, Chekhov S, Zhang K, et al. Sharp wave-ripple disruption limits experience-dependent neuronal ensemble reorganization in amyloid-associated memory impairment. *Manuscript under review at Current Biology*, 2026.

Sun Y, Takehara-Nishiuchi K. The medial prefrontal cortex leaves the hippocampus when it prepares for the future. *Science Progress*, 2024.

Sun Y, Taxis J, Takehara-Nishiuchi K. In silico modeling of amyloidosis-induced CA1 spike abnormalities: Unraveling the role of excess synaptic glutamate. Society for Neuroscience, Poster, 2025.